

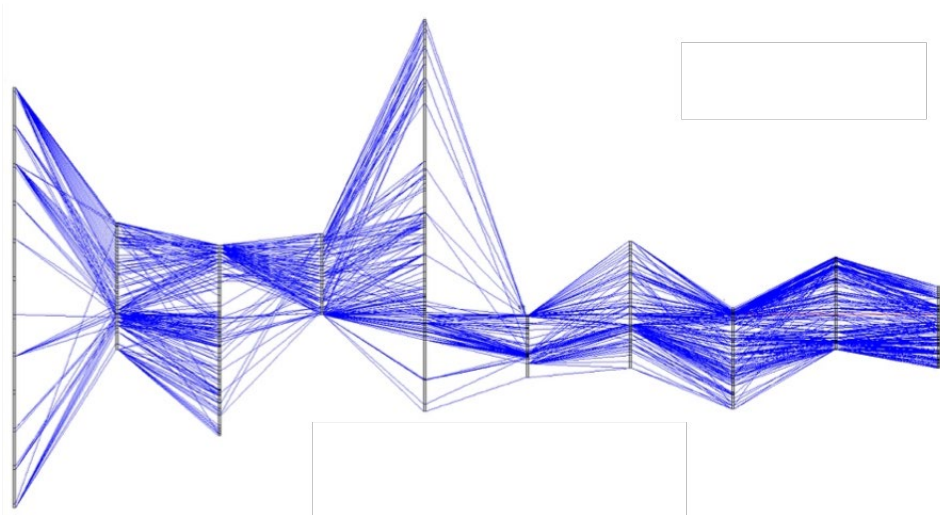
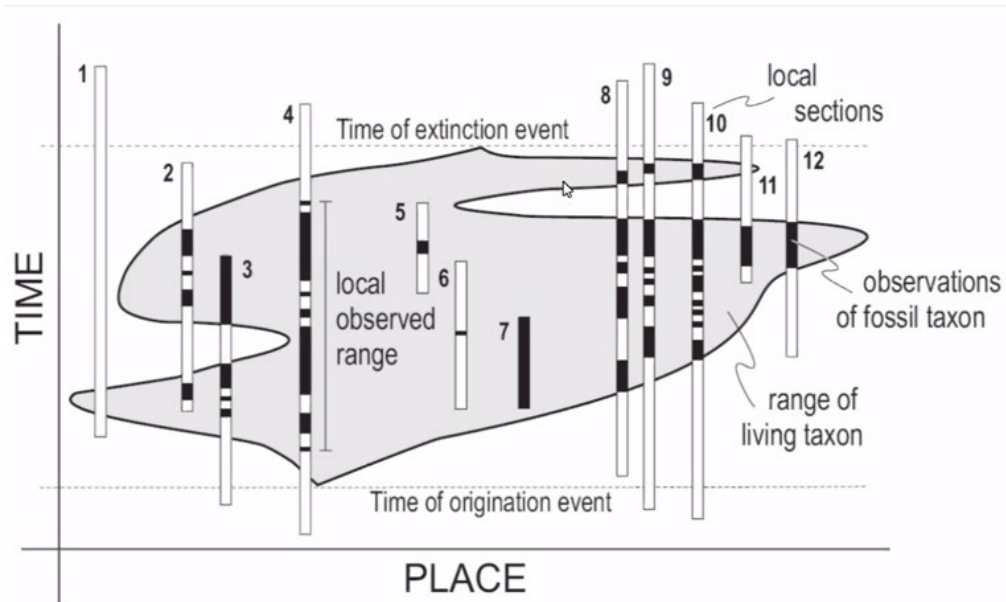


定量地层学与CONOP实验课



鲁铮博 & 施玉洁

地层对比问题引入



- 延限误差：
 - 扩散时差
 - 化石保存不完备
 - 采样不完全

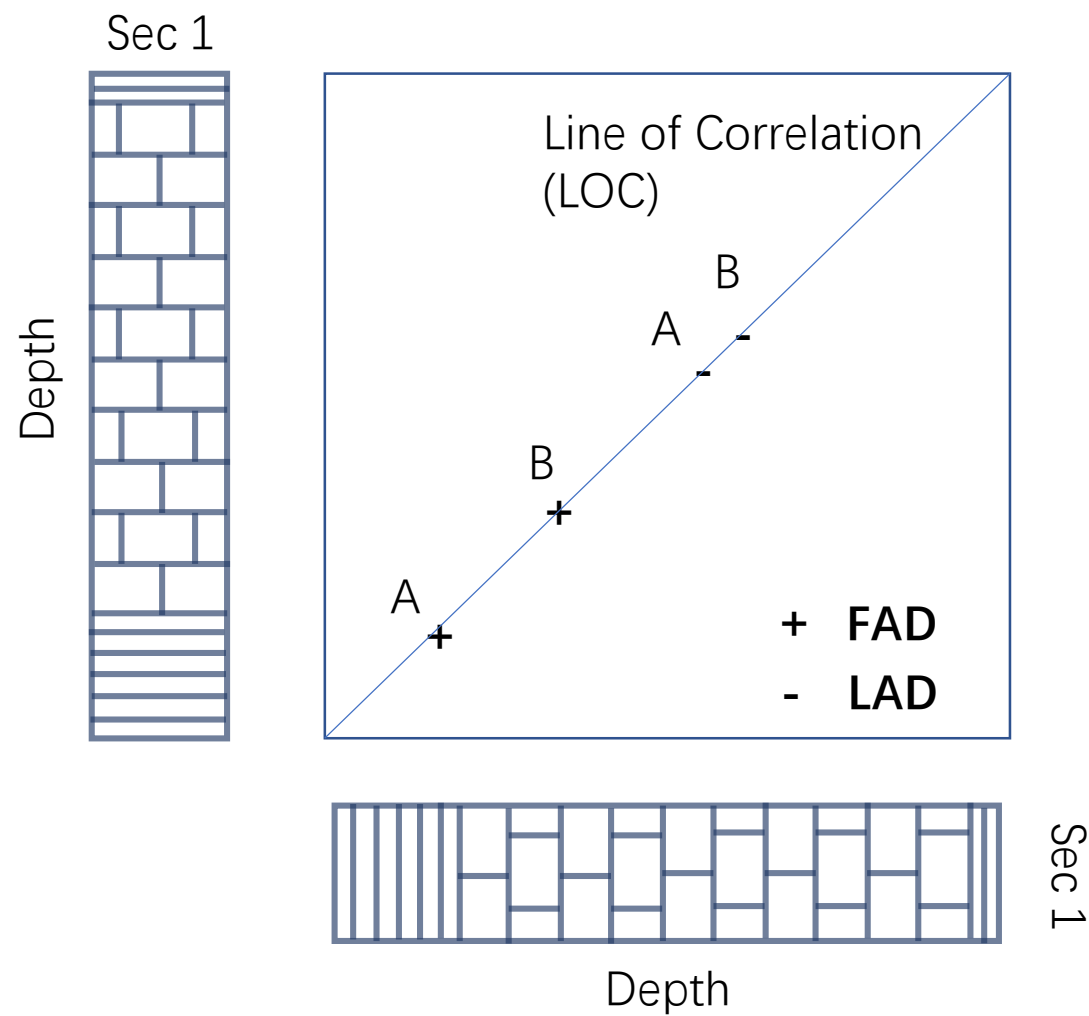


- 化石的真实延限**长于**观测到的延限
- 延限**需要被调整**

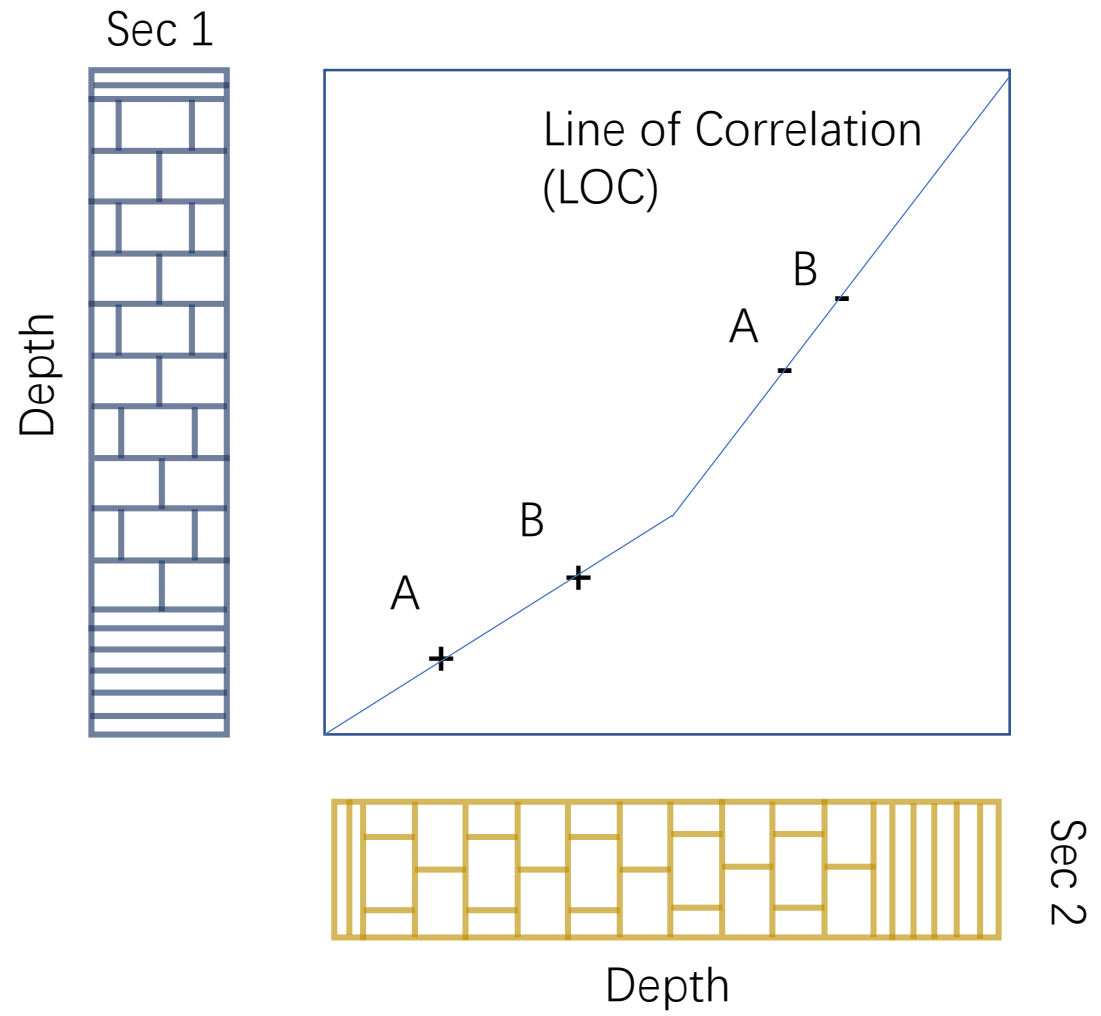
图形对比法——对比线

对比线 (Line of Correlation, 简称LOC): 通常被画在两个或者多个剖面的对比图中, 描述了两个或者多个剖面上地层序列的等时性 (理想的剖面)。

- 对比线必须经过两个剖面间所有等时的事件
- 对比线斜率必须为正



完全等时:
生物事件顺序相同, 所处深度相同



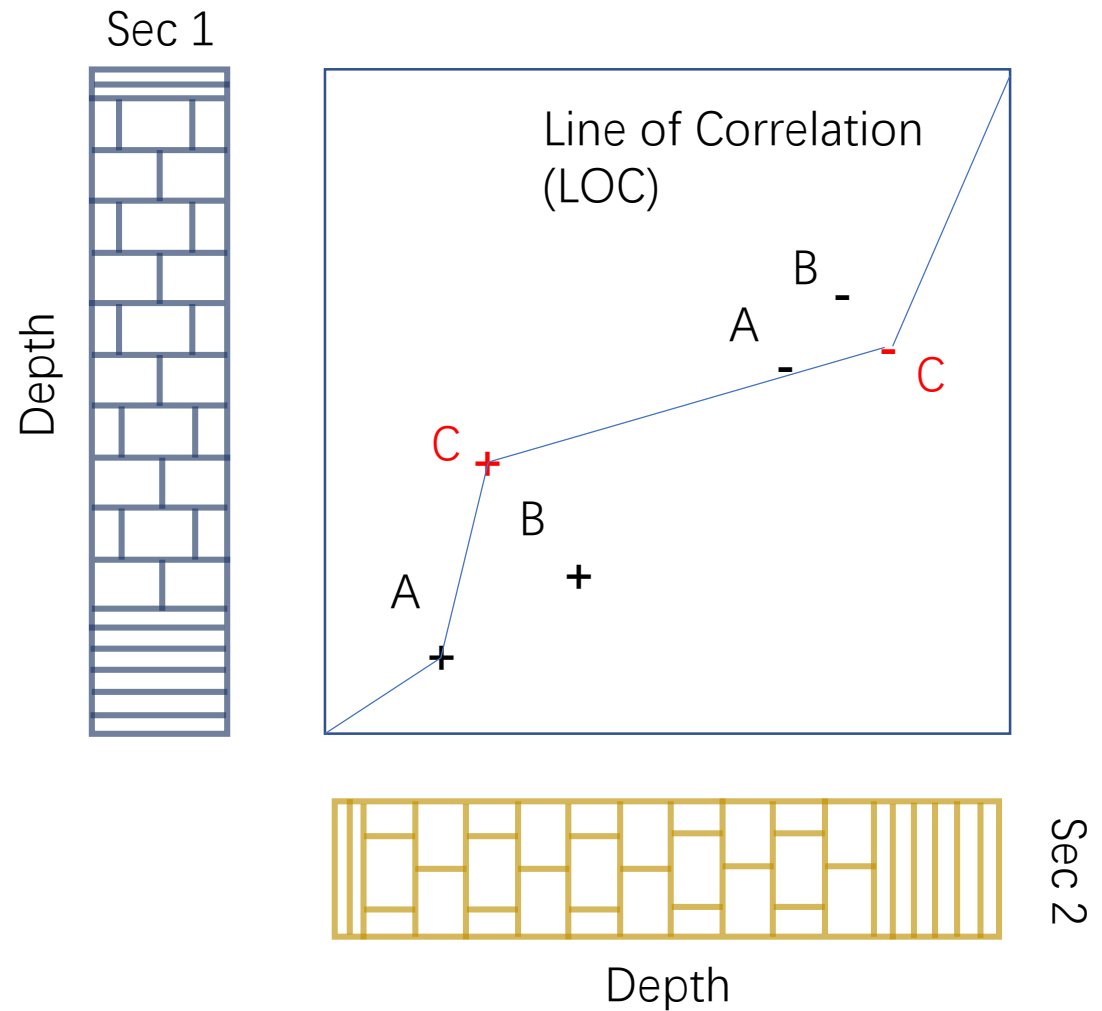
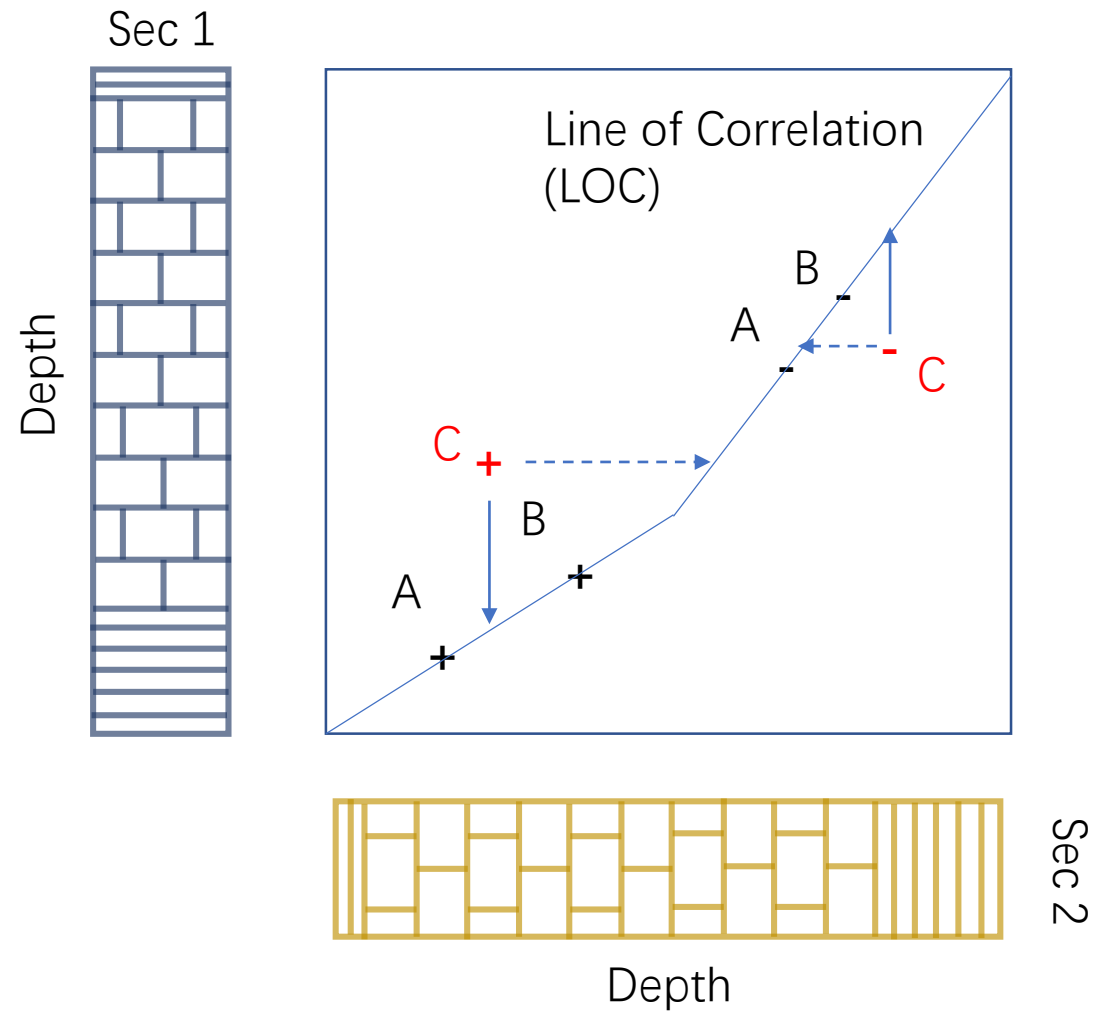
等时但沉积速率的变化:
生物事件顺序相同, 但所处深度不同

图形对比法——事件调整

穿时事件的调整方向存在限制:

- 垂直调整
- 首现向左下角移动, 末现向右上角移动

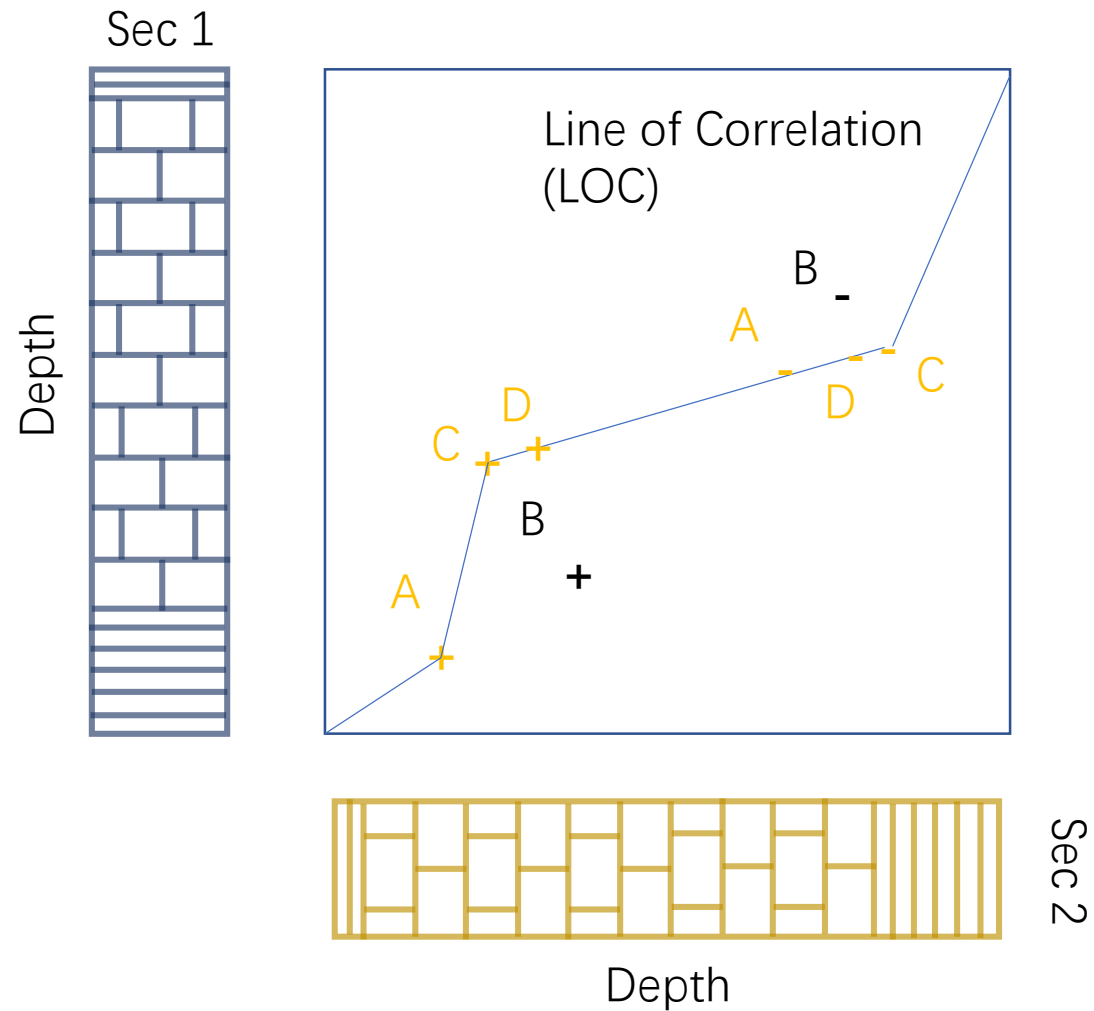
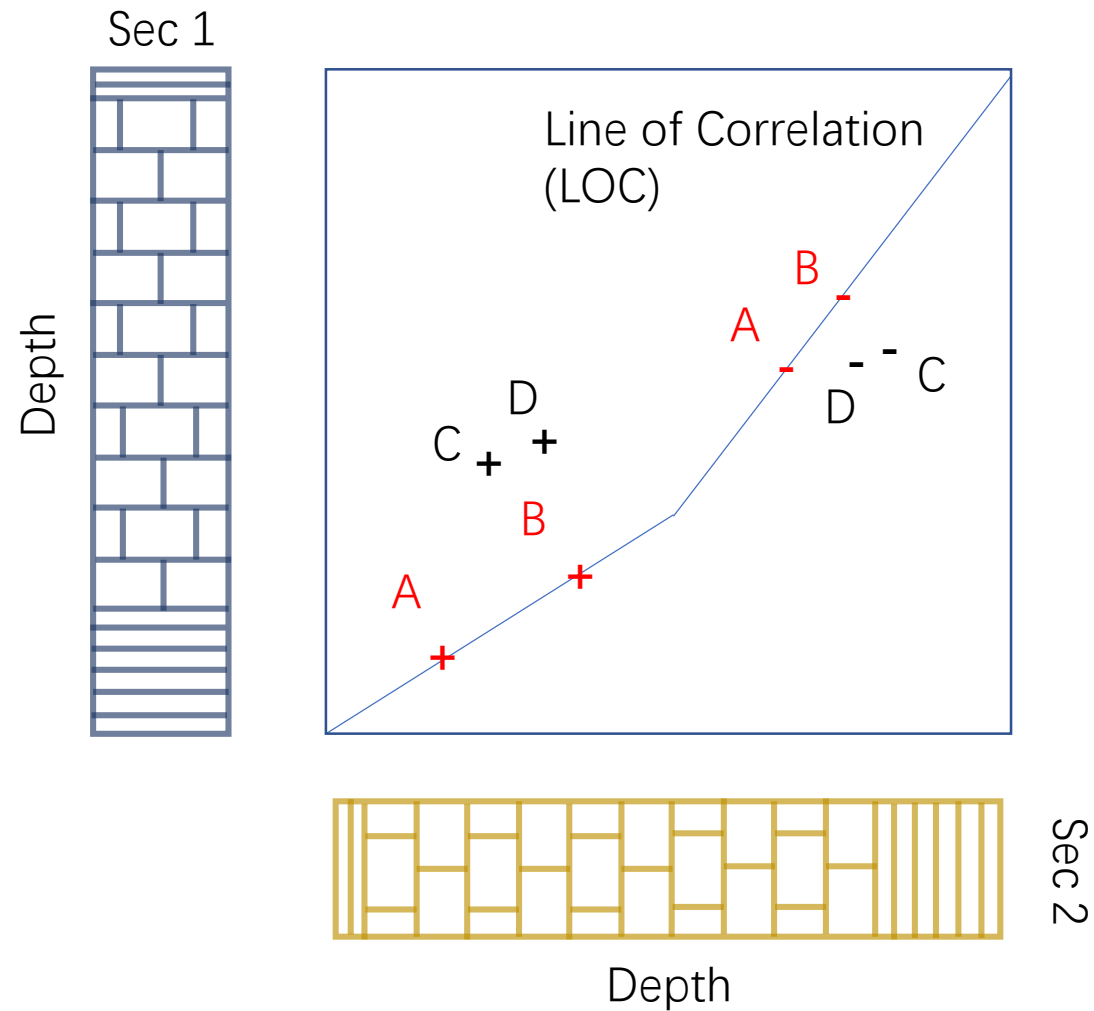
对比线的绘制存在多解性



图形对比法——对比线的多解性

对比线的绘制存在多解性：

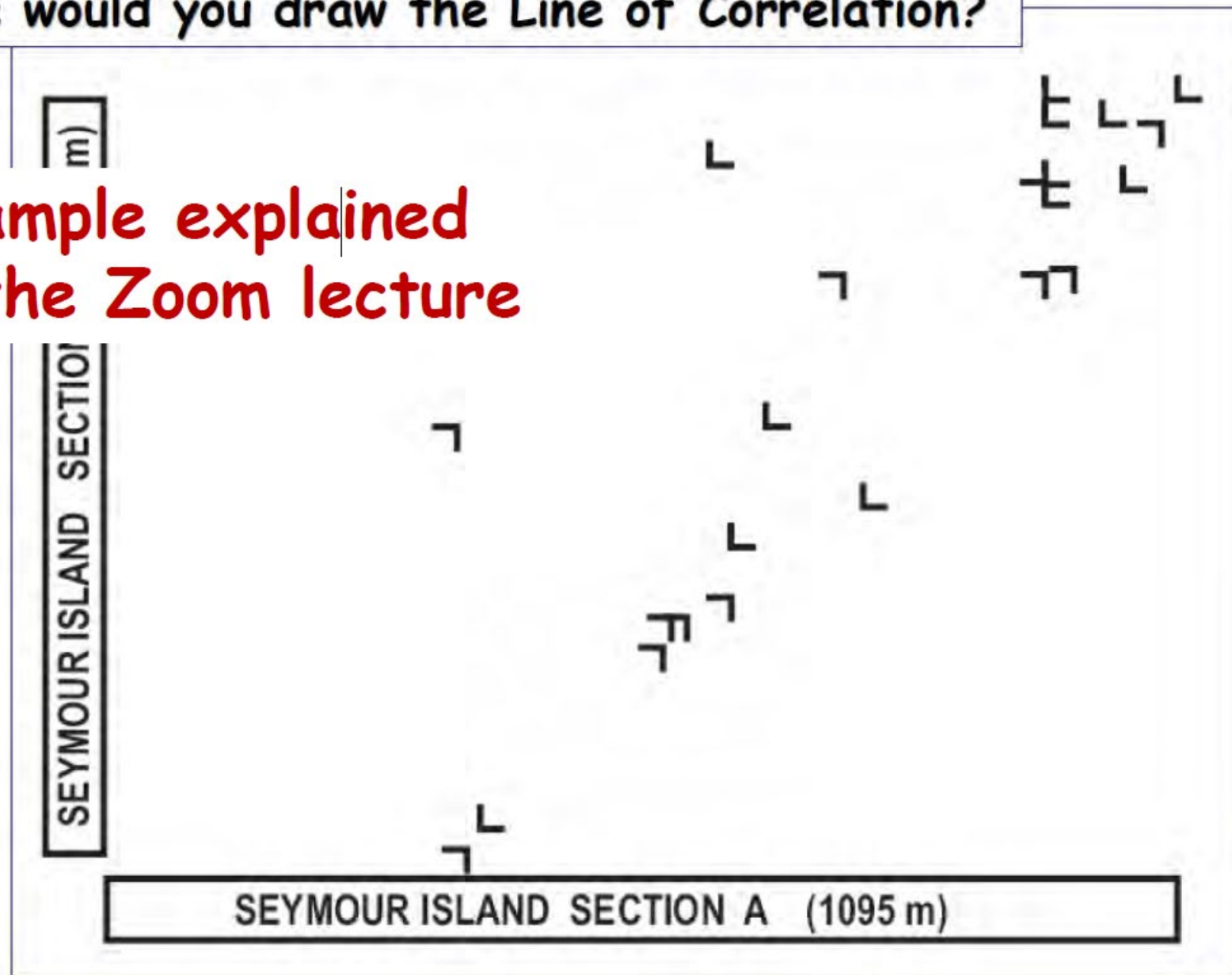
- 专家知识，相信某些化石的等时性
- 让尽可能多的化石存在于LOC上



练习1:

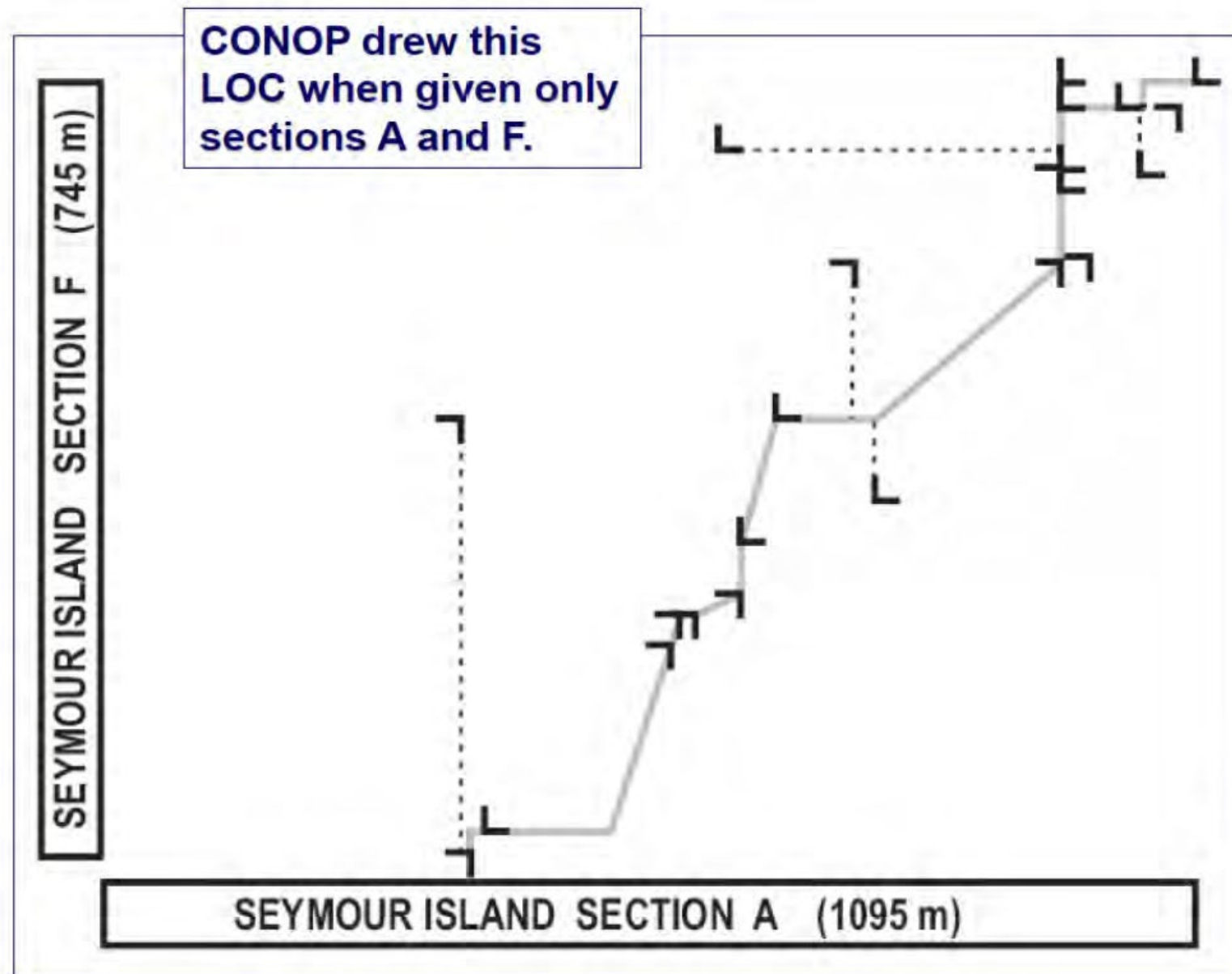
Where would you draw the Line of Correlation?

Example explained
in the Zoom lecture



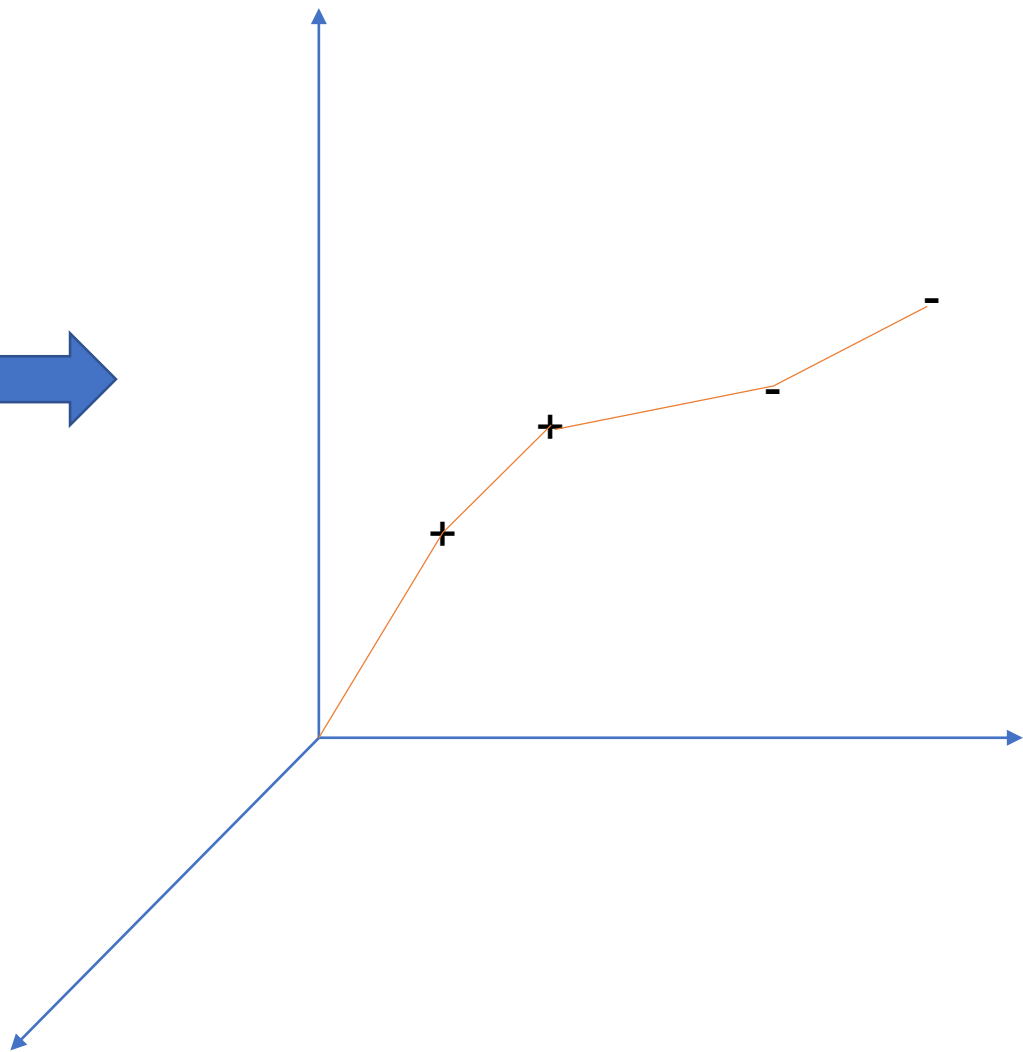
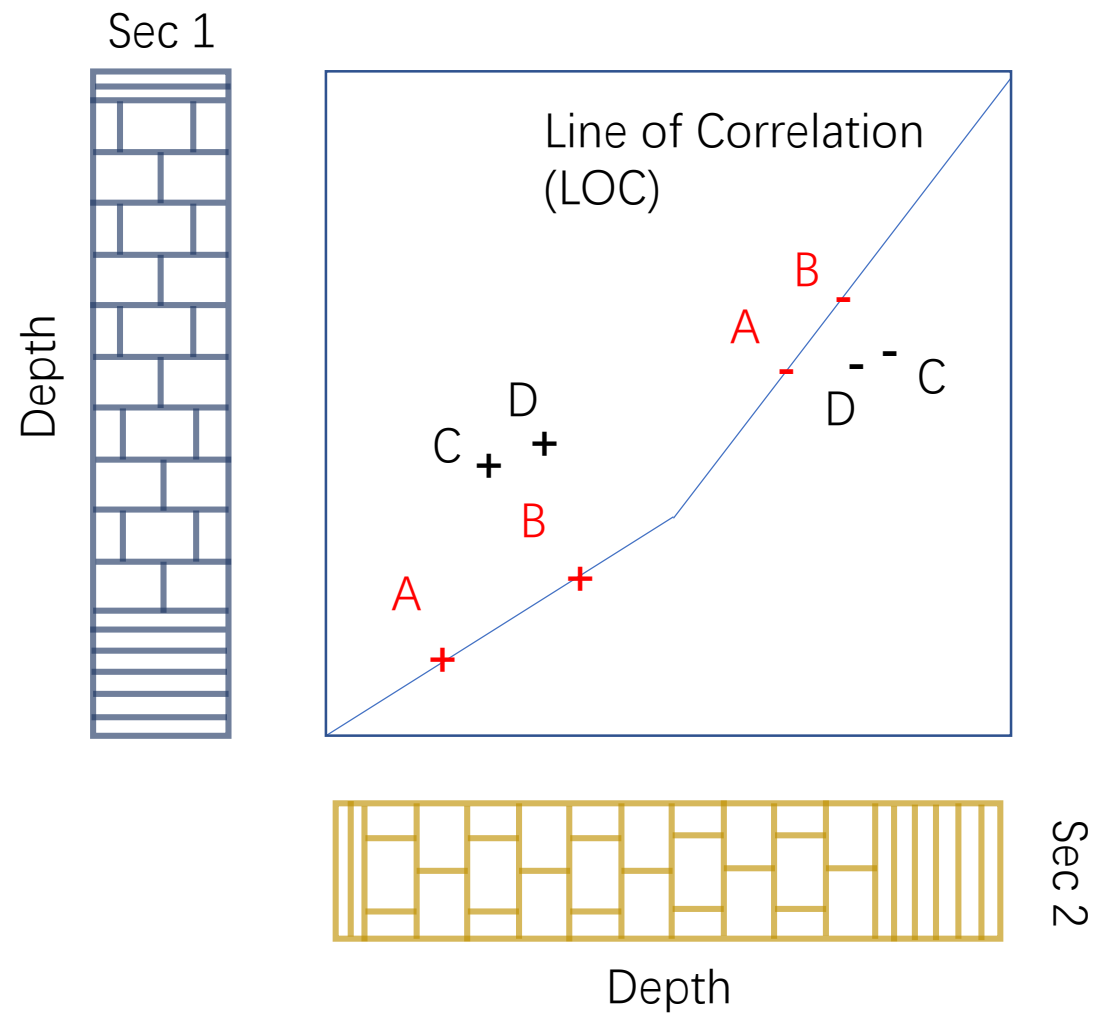
练习1答案:

Graphic Correlation

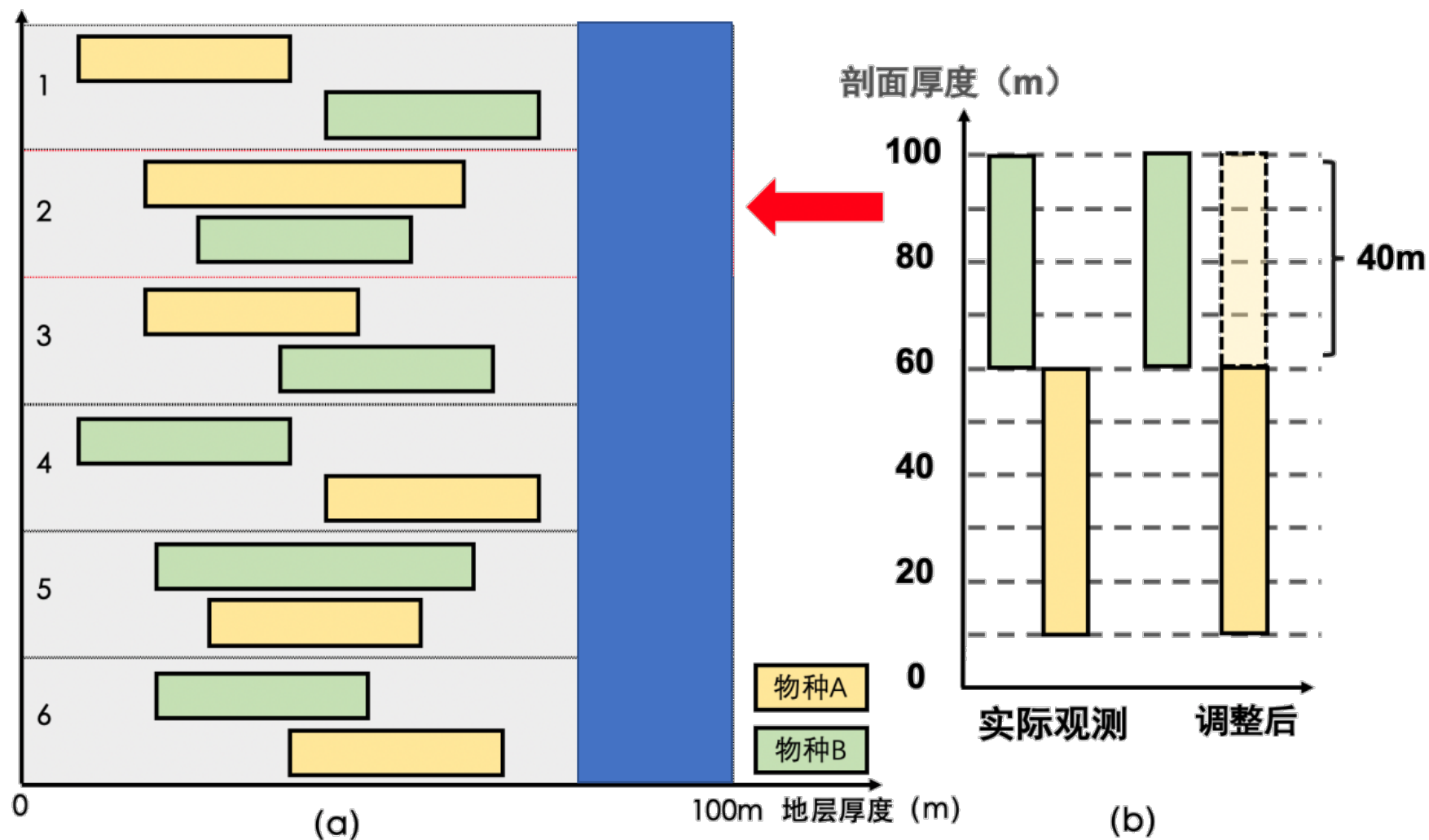


CONOP——n维的图形对比法

3,4,5, ..., n维的图解对比如何实现？

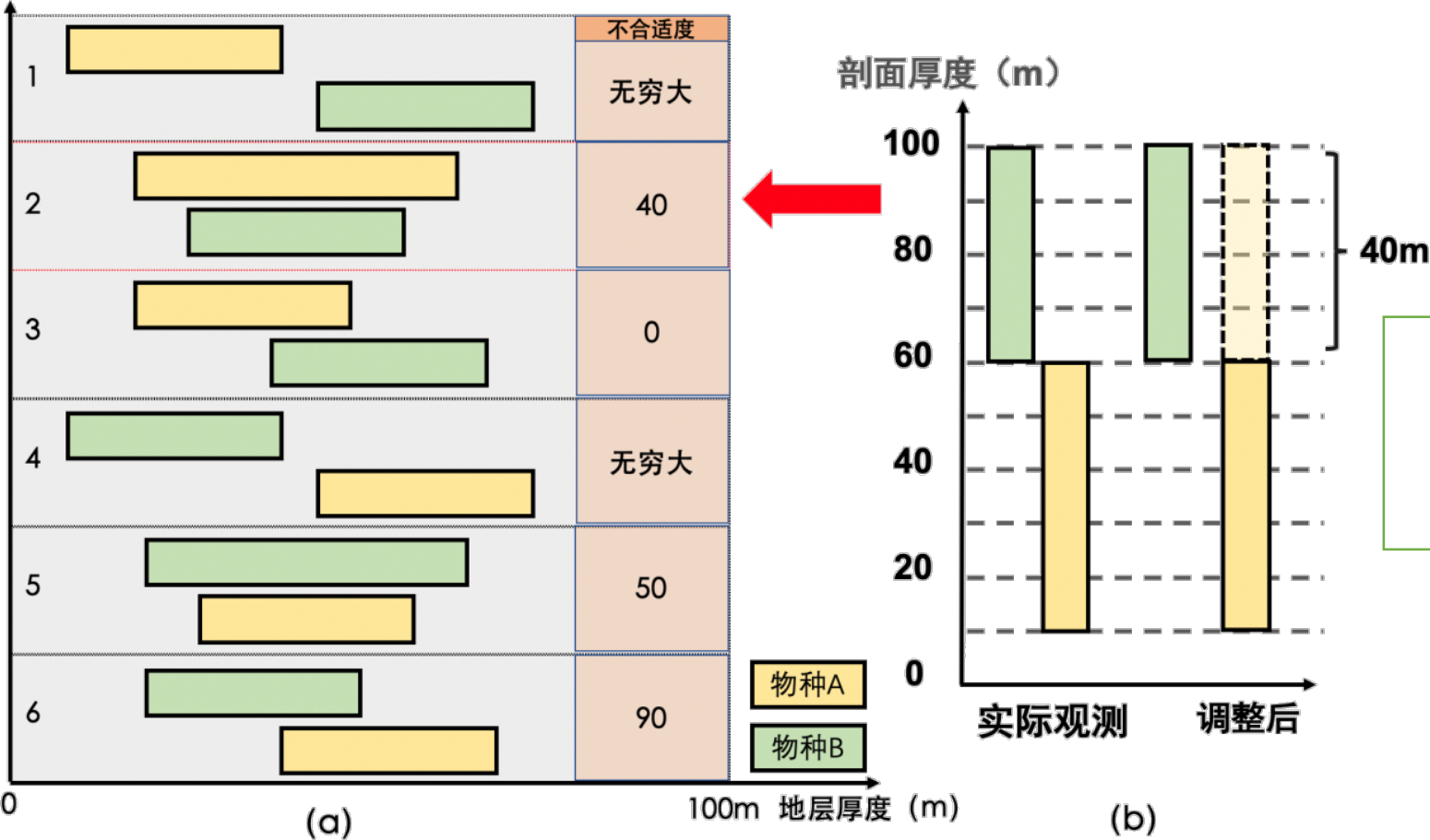


CONOP 原理简介



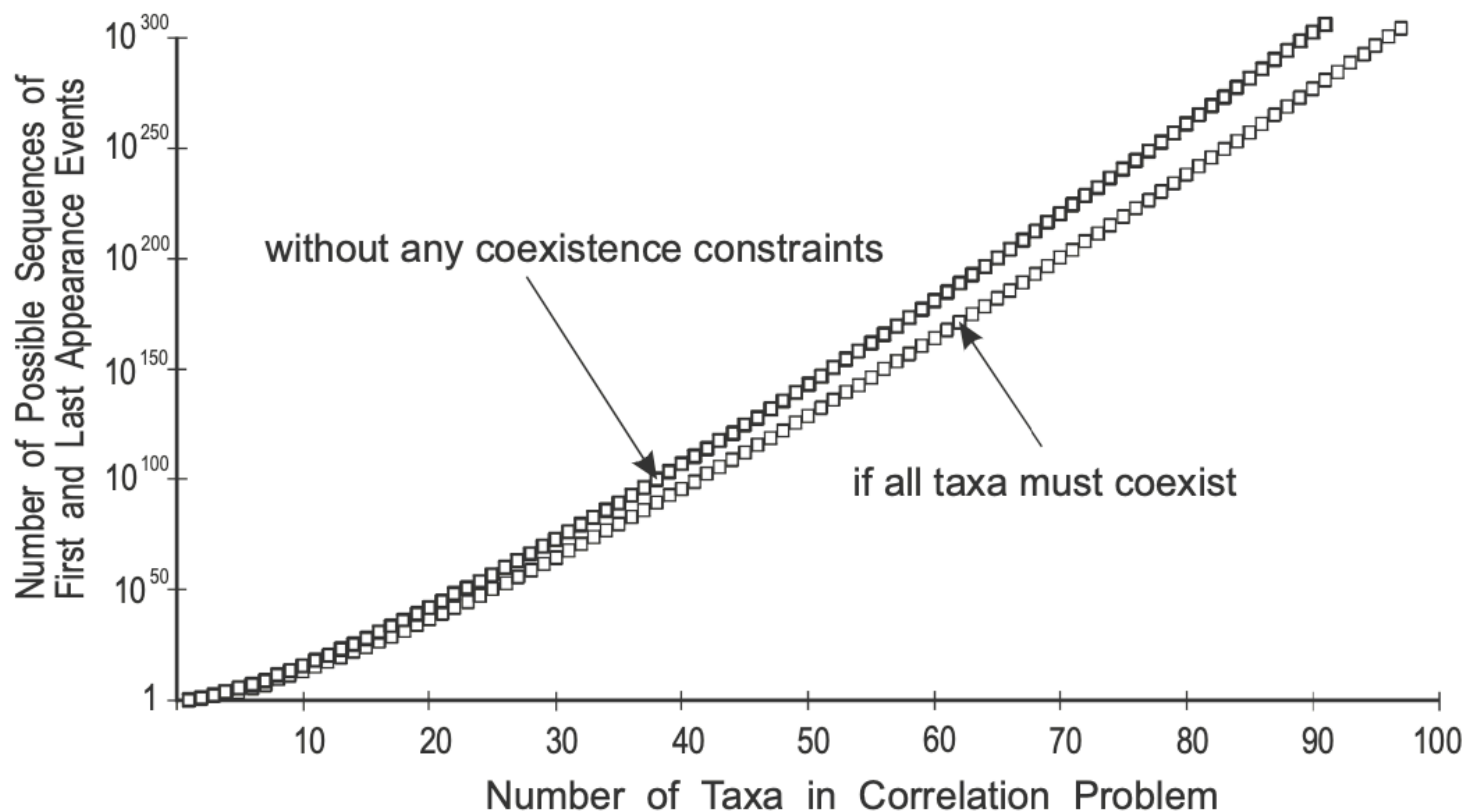
某个化石在被调整延限时，尽可能符合原剖面中和其他化石的延限关系。

CONOP 原理简介



- 可能情况 (N) = 6
- 最小不吻合度 (Penalty) = 40

多剖面多物种情况的推广



- 一个剖面两个物种的情况：
 - $N=6$
 - $\text{Penalty}=40$



推广到m个剖面，n个物种

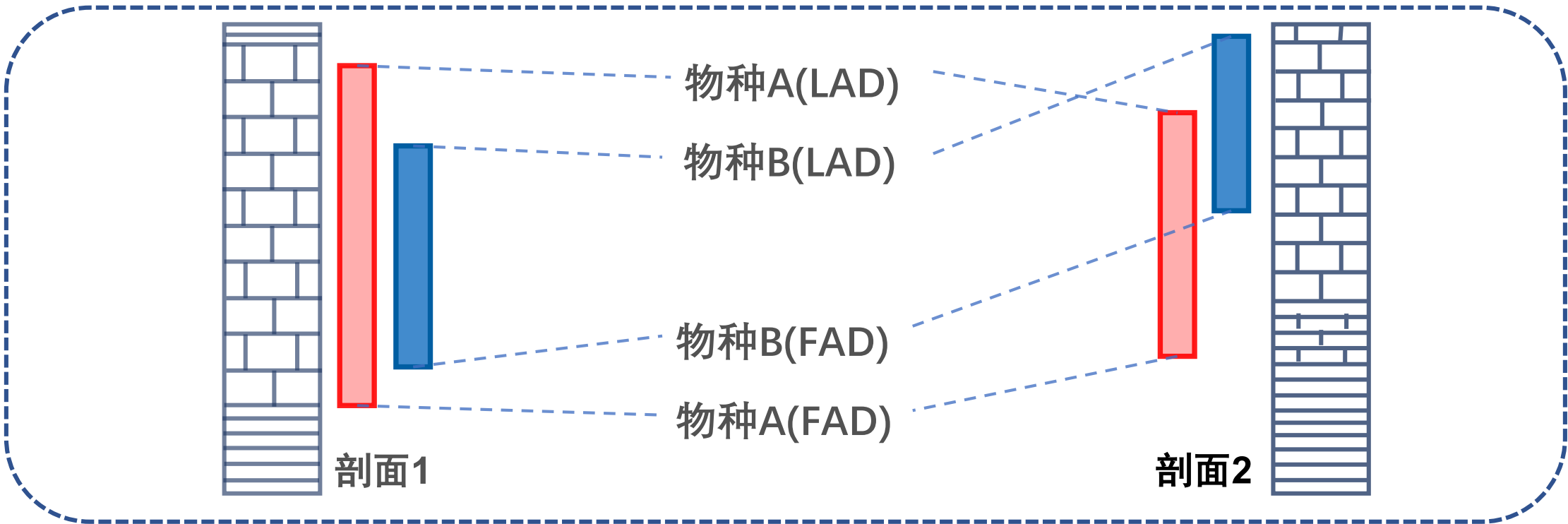
$$N = C_n^1(2n-1) * C_{n-1}^1(2n-3) * C_{n-2}^1(2n-5) * \dots * C_2^1 * 3 * C_1^1 * 1$$

$$\text{penalty} = \sum_{i=1}^n \sum_{j=1}^m [|\alpha_{ij} - a_{ij}| + |\beta_{ij} - b_{ij}|]$$

$\alpha_{ij} > a_{ij}, \beta_{ij} > b_{ij}, \alpha_{ij} > \beta_{ij}$

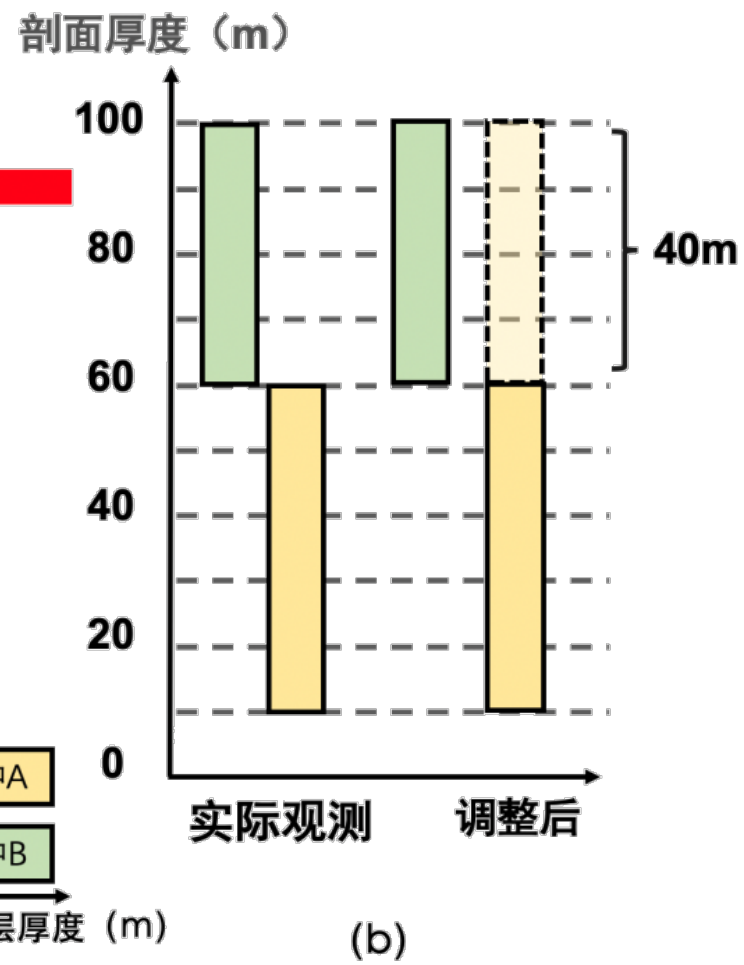
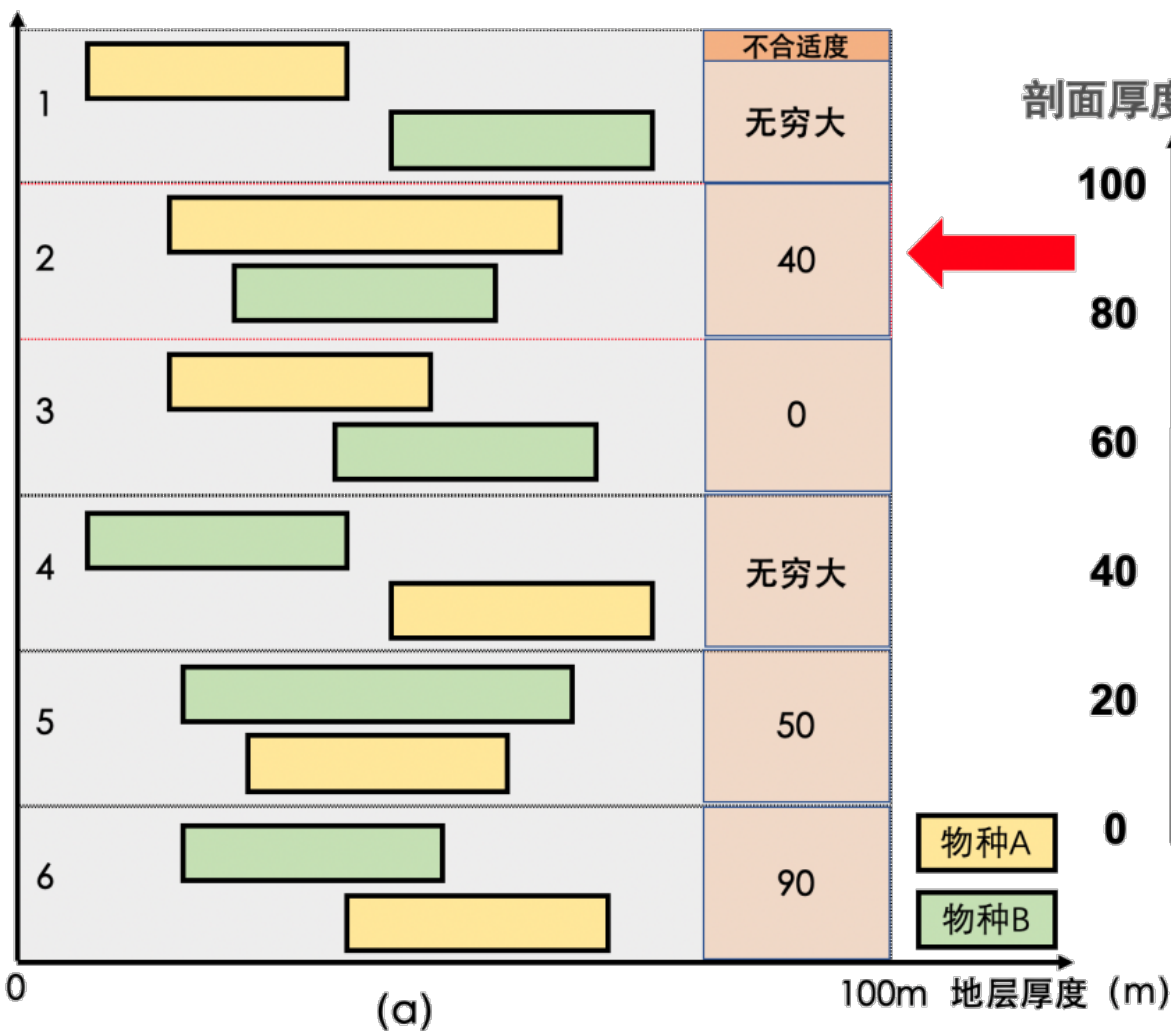
定量地层学问题的数学建模

约束条件：共存关系、首末现原则



(引自邓怡颖)

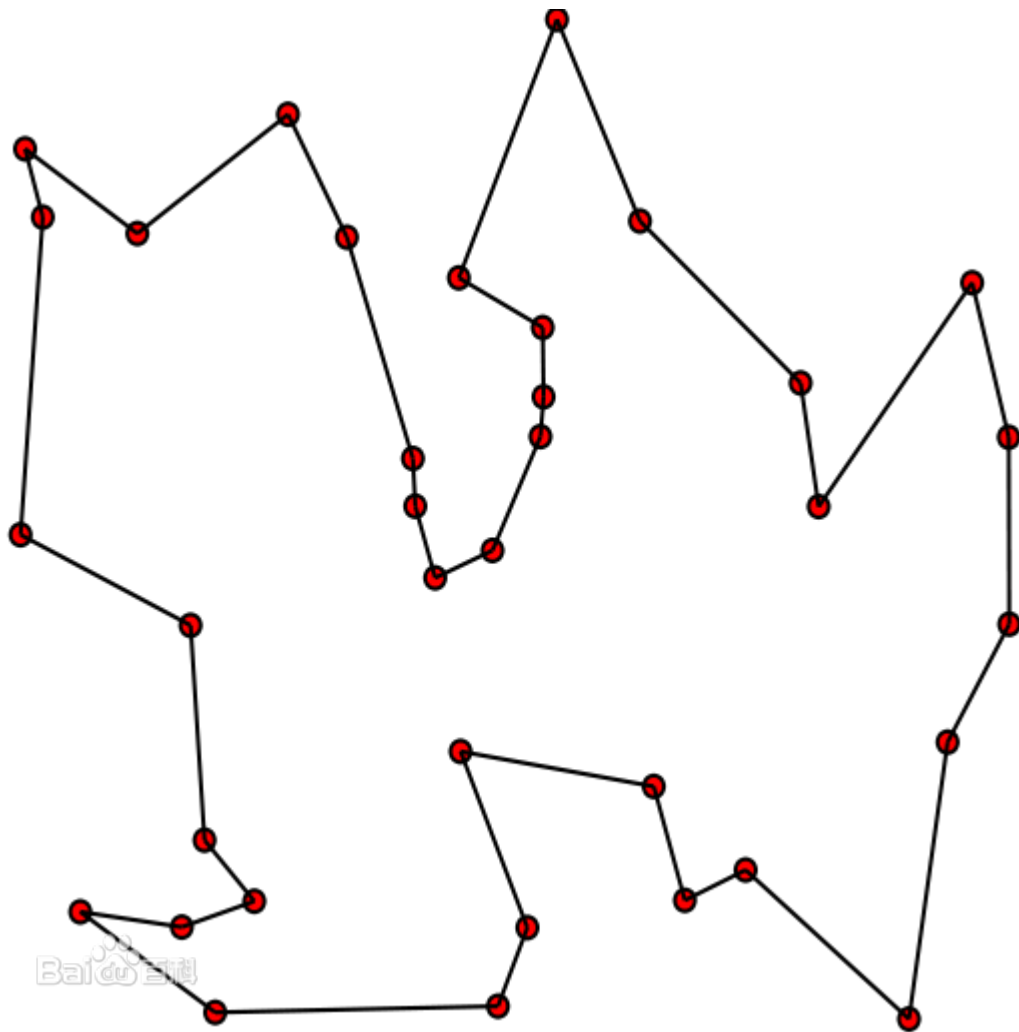
定量地层学问题的数学建模



约束条件：首末现原则、共存关系
优化目标：Penalty

约束最优化问题：在满足一系列有关的限制条件（约束）下，使设计指标（目标）达到最优值。

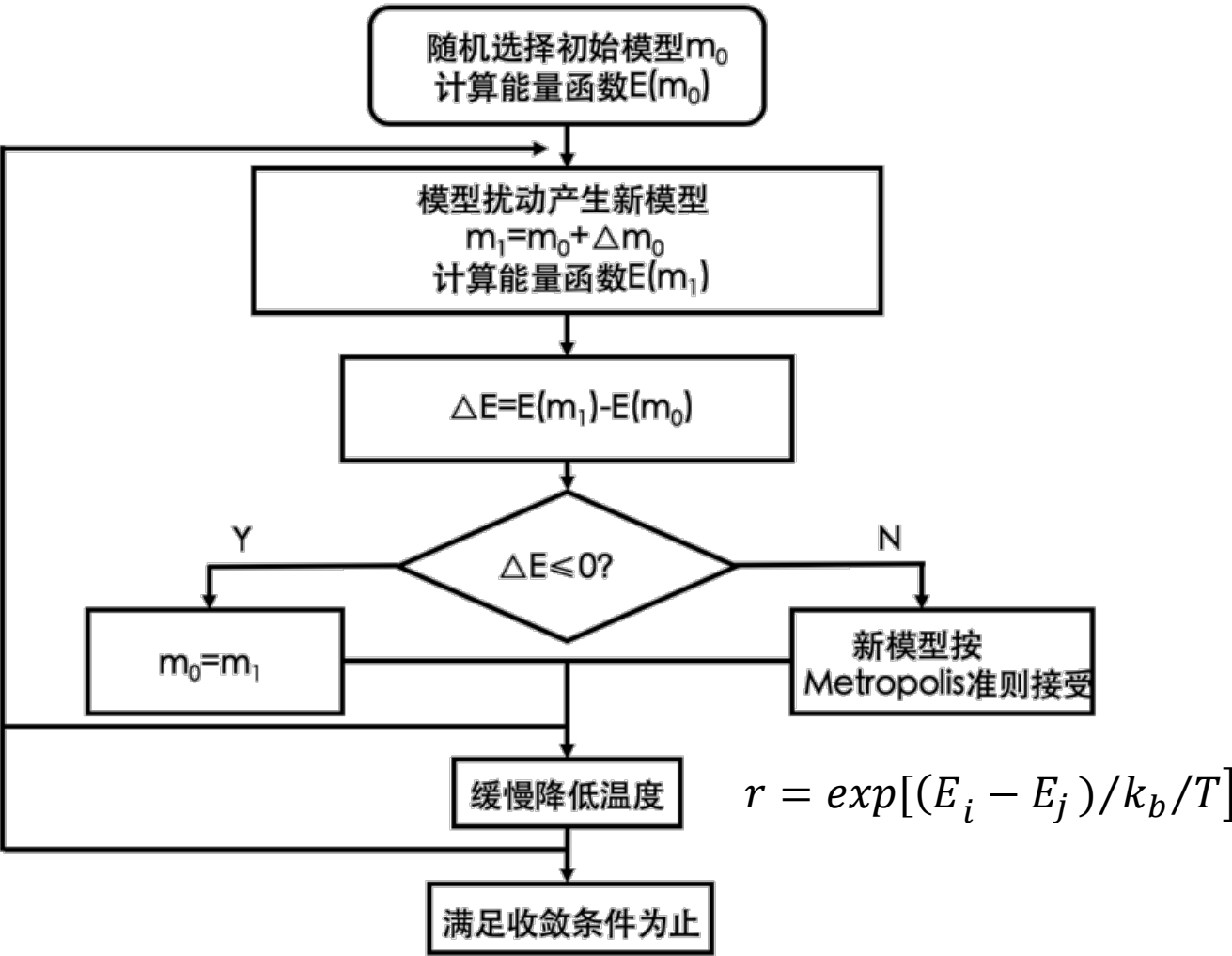
定量地层学问题的数学建模



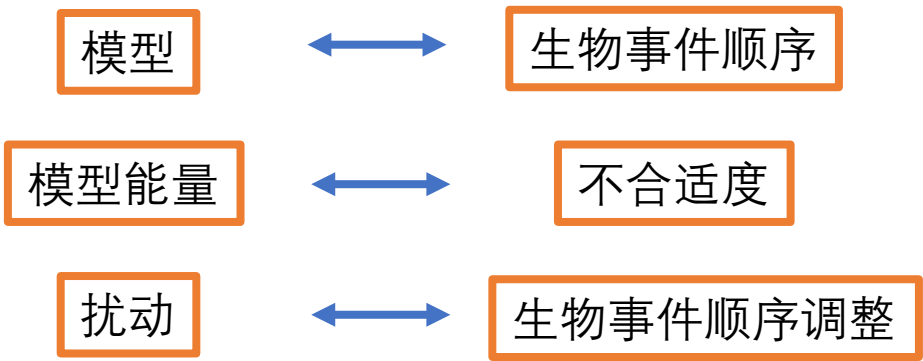
- 旅行推销员问题 (**Travelling salesman problem**, 简称 **TSP**) : 给定一系列城市和每对城市之间的距离, 求解访问每一座城市一次并回到起始城市的最短回路。
- 解法:
 - 精确算法: 分枝定界法、线性规划法等
 - 近似算法或启发式算法: 遗传算法、**模拟退火算法**、贪婪算法和神经网络 (可微) 等

模拟退火算法

模拟退火算法 (Simulated annealing, 简称SA)



地层学问题的算法解决



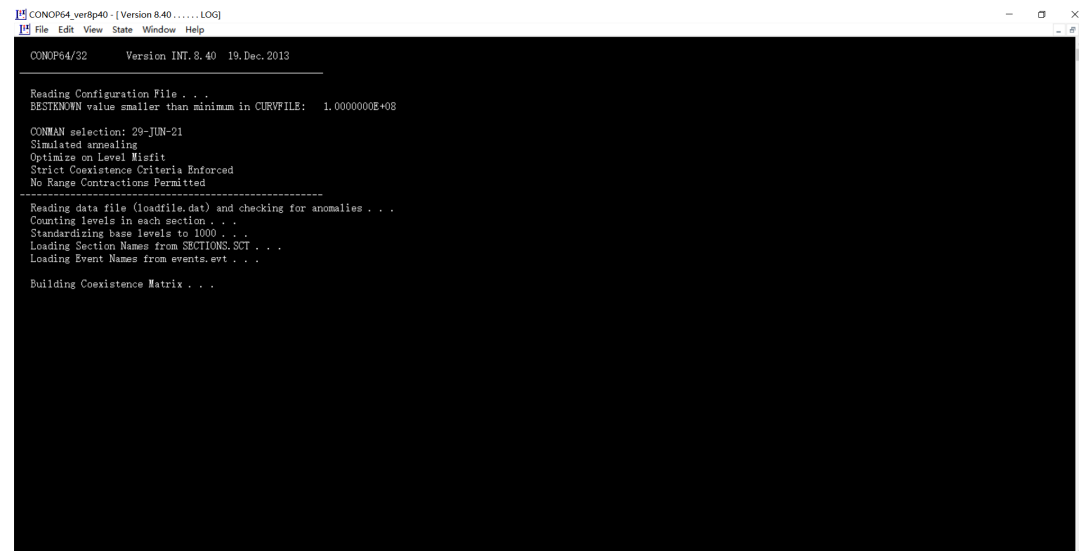
模拟退火算法的参数调整

初始温度 (STARTEMP)、降温比率 (RATIO)、外循环 (STEP)、内循环 (TRIALS)

在实际运算过程中，能否找到最优解（不合适度最小）取决于模拟退火是否具有**足够高的初始温度、缓慢的退火速度、大量的迭代次数以及同一温度下足够的扰动次数**

CONOP的发展

- CONOP最早由Kemple在1995年提出
- 1999年Sadler和Kemple写出了第一个应用程序CONOP
- 2020年，我们团队对CONOP进行了重写，实现了并行化运算，使得其运行速度提升了10倍以上

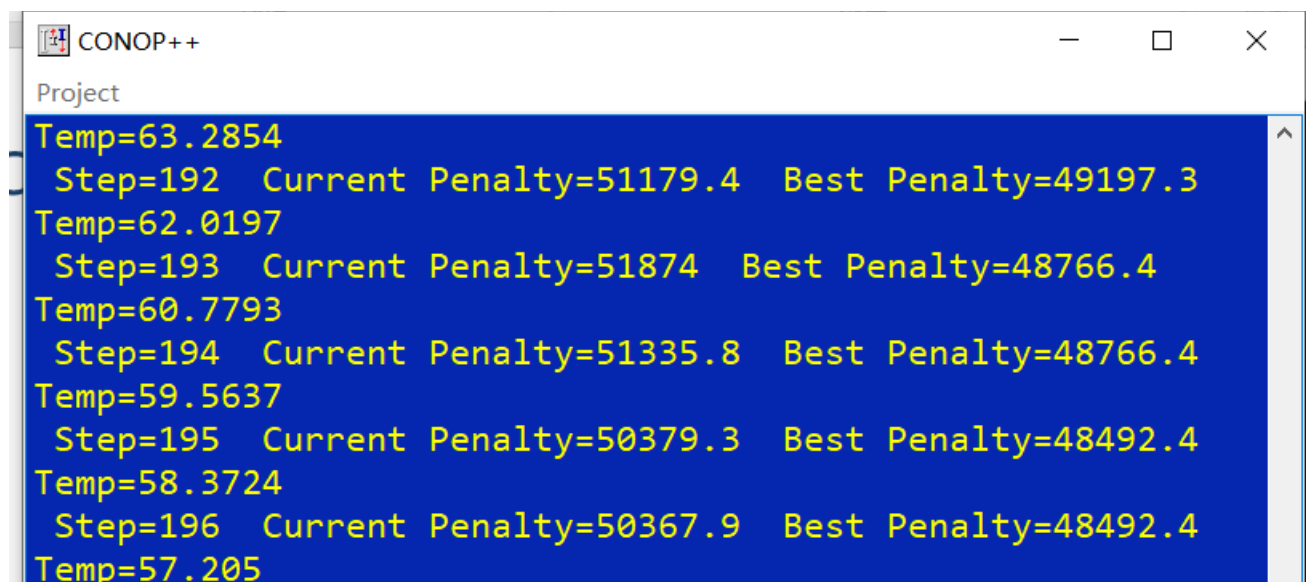


```
CONOP64/32      Version INT. 8.40  19.Dec.2013

Reading Configuration File . . .
BESTKNOWN value smaller than minimum in CURVFILE:  1.0000000E+08

CONMAN selection: 29-JUN-21
Simulated annealing
Optimize on Level Misfit
Strict Coexistence Criteria Enforced
No Range Contractions Permitted
-----
Reading data file (loadfile.dat) and checking for anomalies . . .
Counting levels in each section . . .
Standardizing base levels to 1000 . . .
Loading Section Names from SECTIONS.SCT . . .
Loading Event Names from events.evt . . .
Building Coexistence Matrix . . .
```

CONOP64运行界面（Sadler & Kemple编写）

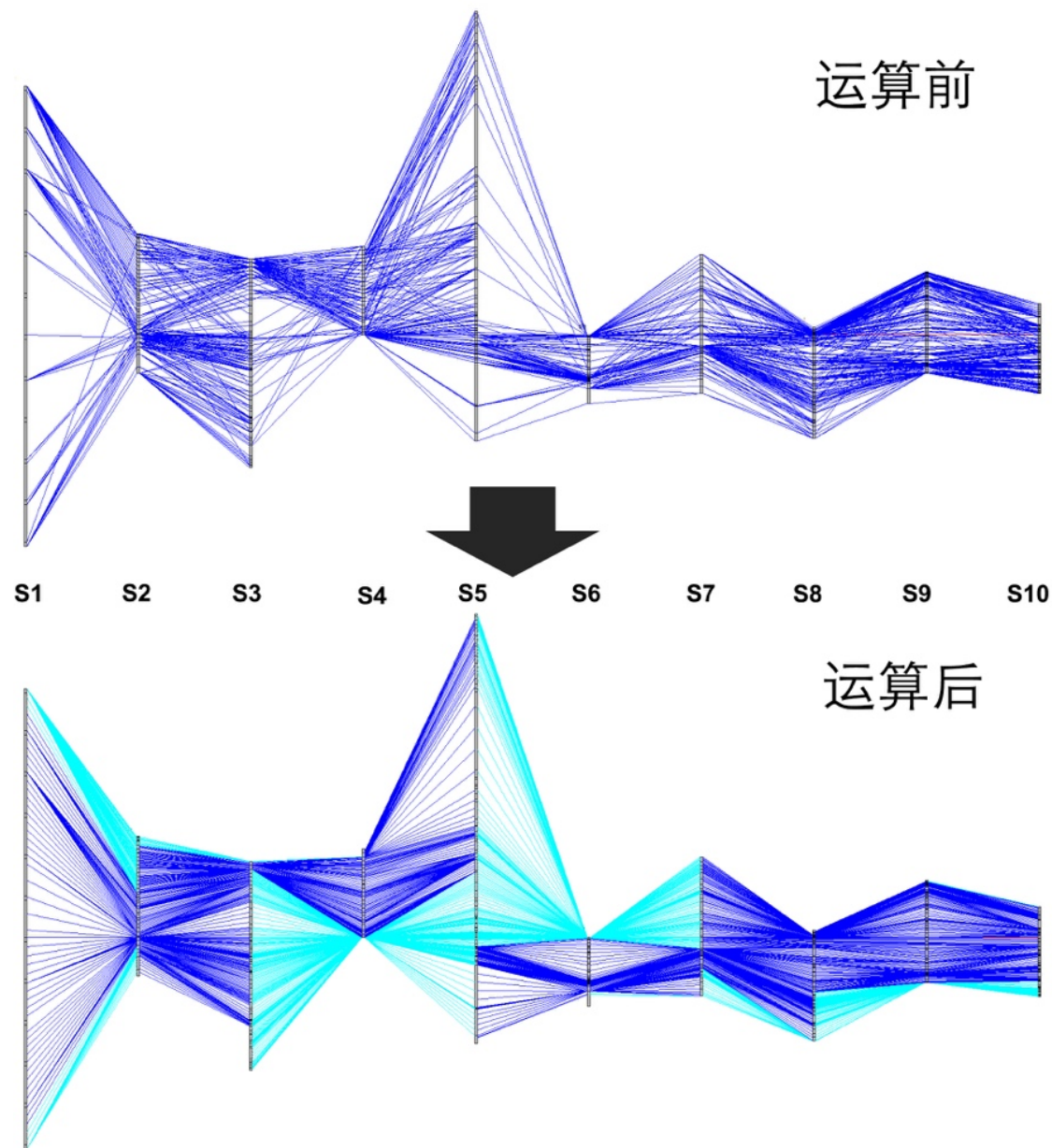


```
CONOP++
Project
Temp=63.2854
Step=192  Current Penalty=51179.4  Best Penalty=49197.3
Temp=62.0197
Step=193  Current Penalty=51874  Best Penalty=48766.4
Temp=60.7793
Step=194  Current Penalty=51335.8  Best Penalty=48766.4
Temp=59.5637
Step=195  Current Penalty=50379.3  Best Penalty=48492.4
Temp=58.3724
Step=196  Current Penalty=50367.9  Best Penalty=48492.4
Temp=57.205
```

CONOP2020 运行界面（侯旭东编写）

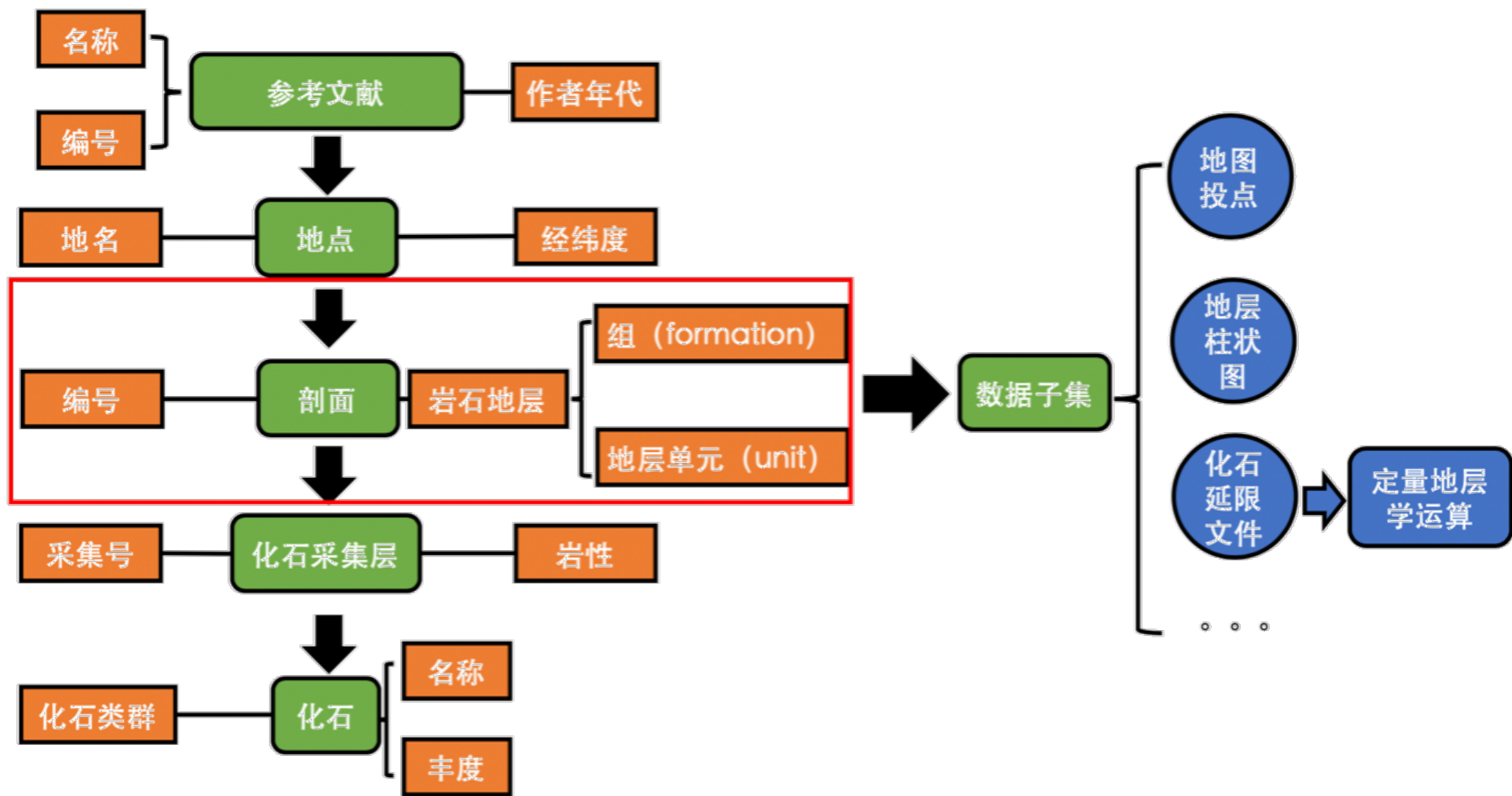
CONOP的优势

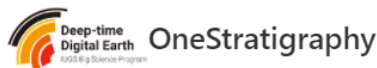
- CONOP作为定量地层学方法的一种，可以保留所有生物事件用于地层划分与对比，时间分辨率可以比传统方法提高数百倍。
- CONOP方法实现了地层对比的自动化。其相较于传统定量地层学方法（图形对比法）速度更快、效率更高。
- 借助并行计算和超级计算机，CONOP方法的运算速度可以进一步提升。



OneStratigraphy 数据库简介

OneStratigraphy数据库：是一个为地层学信息的数字化、标准化、集成、共享和使用而专门设计的平台，由南京大学的樊隽轩等在2019年开始设计、建设并逐步对外开放。





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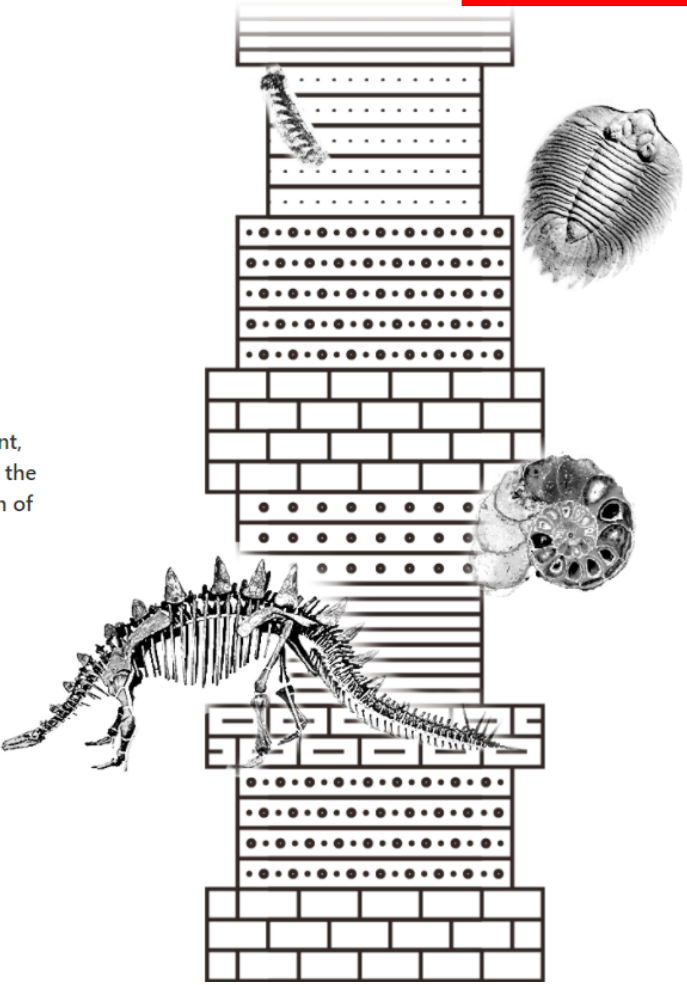
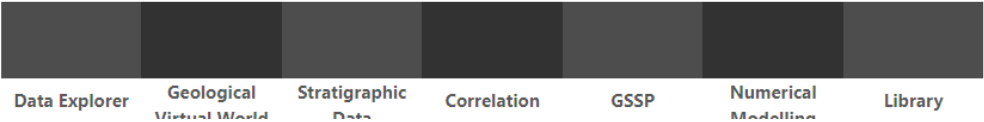
recophin



OneStratigraphy

OneStratigraphy is a community-based, international collaboration project. It focuses on the integration, management, sharing, visualization and analytics of stratigraphic data. 'One' means one consistent stratigraphic framework around the world, which helps understanding the billions of Earth history, regional and global stratigraphic correlation, evolution of life etc.

View more





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Data Explorer

Geological
Visual World

Stratigraphic
Data

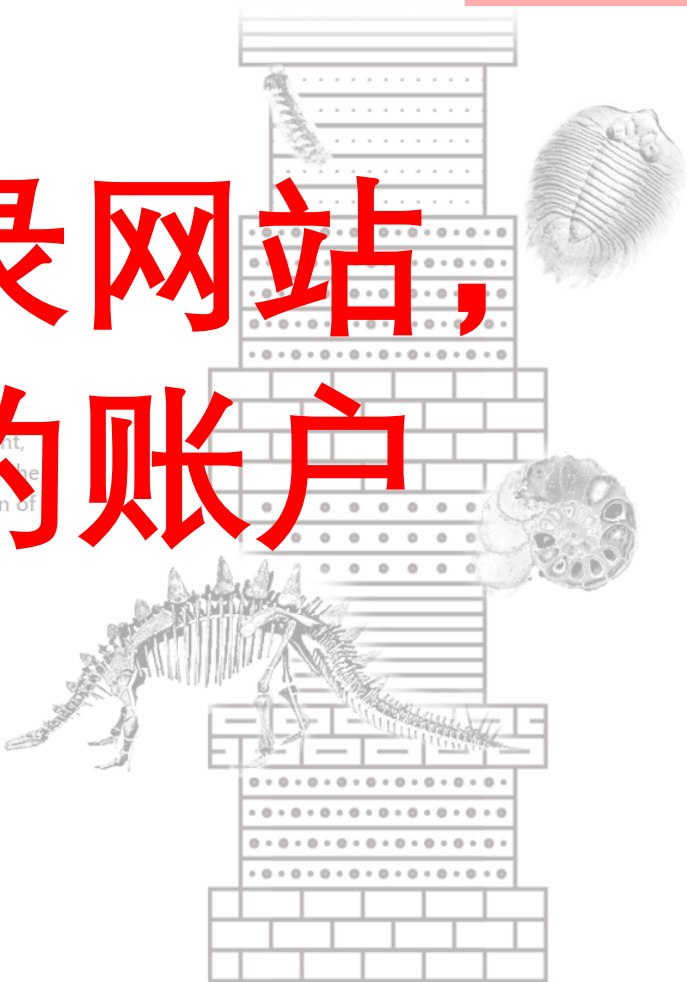
Correlation

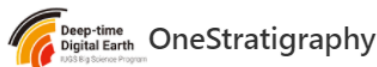
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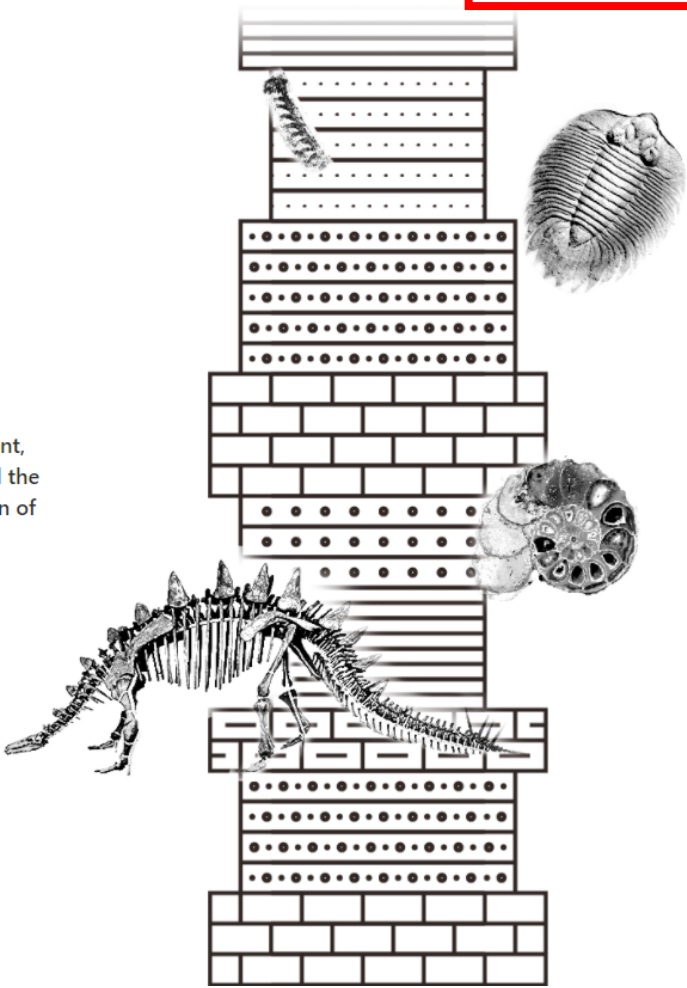
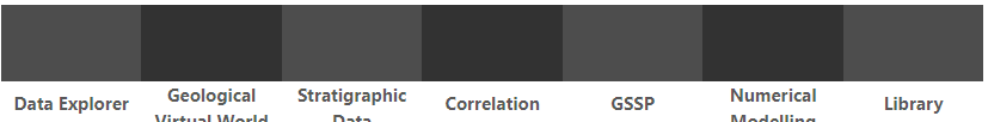
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序号	剖面名称	参考文献	审核人	数据编号
1	Varignano	Luciani et al., 2020		
2				SE202106300741452
3	Massignano	Molina et al., 1993		
4		Firth et al., 2012	Lu Zhengbo	
5	Agost		Lu Zhengbo	
6	Site 270	Kulhanek et al., 2019		
7	Alano	Agnini et al., 2021		
8		Gennari et al., 2018	Lu Zhengbo	
9	Contessa Barbetti Road (CBR)	Coccioni et al., 2008		
10	Kysing-4			

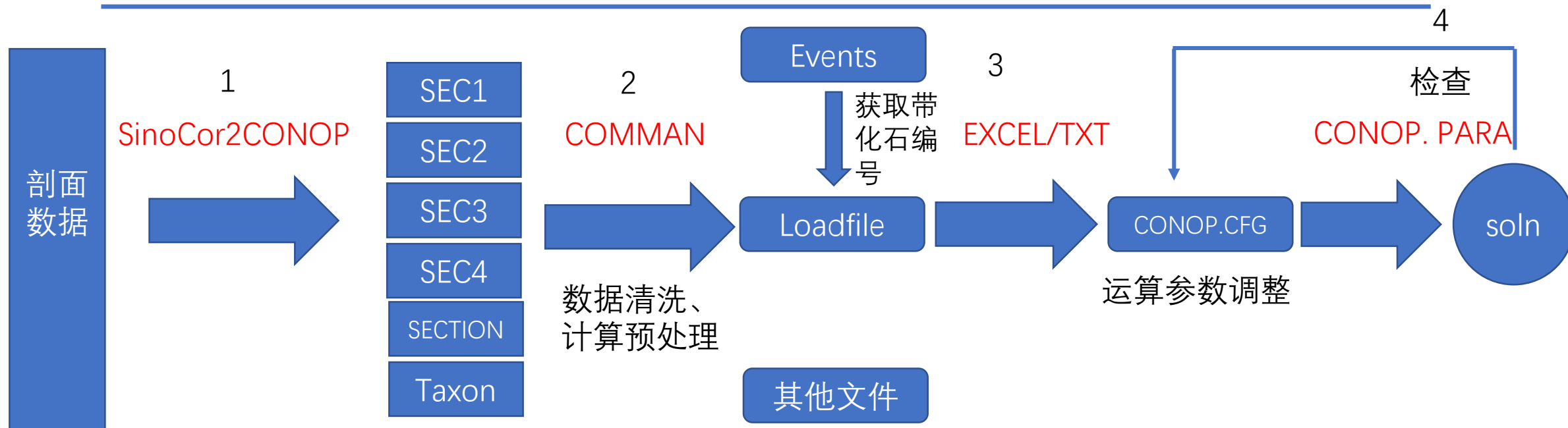
CONOP运算流程

SinoCor文件

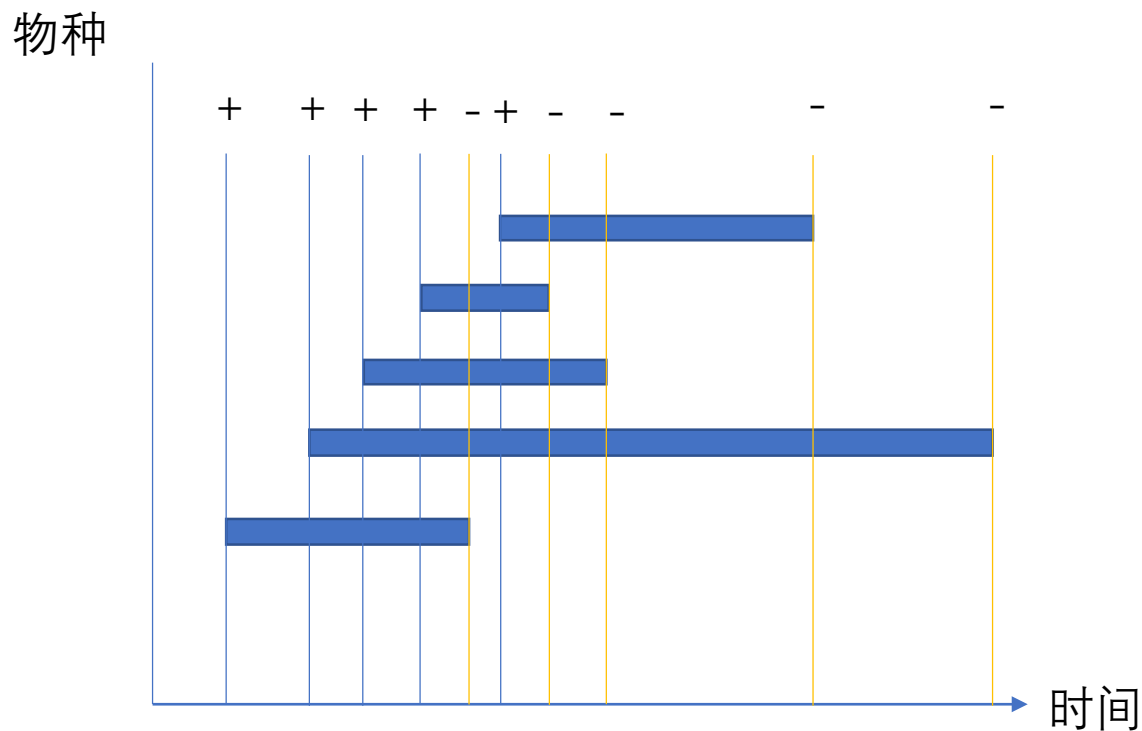
CONOP文件

CONOP运算文件

CONOP结果文件



非时间窗（unbin）生物多样性分析



生物多样性/物种分异度计算：非时间窗方法假定CONOP运算获得的每一次生物事件（首现或未现事件）都代表了一次新生或灭绝事件，并且事件间没有明显的物种演替（Cooper et al., 2014）。每一次新生或灭绝事件都可以被视为一个界线，在界线处如果发生新生事件则给物种数量加1，如果发生灭绝事件则给物种数量减1，最终使用物种数量指示物种多样性。

谢谢！

